

## PCF 2.0 — Develop and Manage Products and Services

HITL Maturity Assessment — All Processes

This document reports HITL (Human-in-the-Loop) maturity ratios for all of PCF 2.0 (Develop and Manage Products and Services) — 3 process groups, 12 Level 3 processes, and 65 Level 4 activities in total — at five maturity levels from M1 (Undesigned) through M5 (Leading), with M3 marking HITL Design Intent.

At M3 (HITL Design Intent), this domain's aggregate human-involvement ratio is **59%**, down from a **80%** undesigned (M1) baseline — a **-21-point** AI convergence opportunity. Mixed-pattern L3 rows reflect an equal-weighted rollup of their real L4 children rather than a force-classified single pattern.

### HITL Maturity Assessment

H% = Human involvement at each maturity level · M3 = HITL Design Intent · Full L1→L2→L3→L4 hierarchy

| PCF     | L1-Domain / L2-Process Group / L3-Process / L4-Activity                         | Pattern                                                                             | M1  | M2  | M3 ★ | M4  | M5  | Δ M1→M3 |
|---------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----|-----|------|-----|-----|---------|
| 2.0     | Develop and Manage Products and Services · L1 DOMAIN AGGREGATE                  | —                                                                                   | 80% | 71% | 59%  | 46% | 36% | -21%    |
| 2.1     | Govern and manage product/service development program · L2 AVG                  | —                                                                                   | 78% | 70% | 58%  | 46% | 36% | -20%    |
| 2.1.1   | Manage product and service portfolio                                            |    | 87% | 81% | 70%  | 59% | 50% | -17%    |
| 2.1.1.1 | Evaluate performance of existing products/services against market opportunities | Knowledge                                                                           | 85% | 78% | 65%  | 52% | 42% | -20%    |
| 2.1.1.2 | Confirm alignment of product/service concepts with business strategy            | Decision                                                                            | 92% | 88% | 80%  | 72% | 65% | -12%    |
| 2.1.1.3 | Prioritize and select new product/service concepts                              | Decision                                                                            | 92% | 88% | 80%  | 72% | 65% | -12%    |
| 2.1.1.4 | Plan and develop cost and quality targets                                       | Knowledge                                                                           | 85% | 78% | 65%  | 52% | 42% | -20%    |
| 2.1.1.5 | Specify development timing targets                                              | Knowledge                                                                           | 85% | 78% | 65%  | 52% | 42% | -20%    |
| 2.1.1.6 | Plan for product/service offering modifications                                 | Knowledge                                                                           | 85% | 78% | 65%  | 52% | 42% | -20%    |
| 2.1.2   | Manage product and service life cycle                                           |  | 75% | 66% | 53%  | 40% | 30% | -22%    |
| 2.1.2.1 | Develop plan for new product/service development and introduction/launch        | Knowledge                                                                           | 85% | 78% | 65%  | 52% | 42% | -20%    |
| 2.1.2.2 | Introduce new products/services                                                 | Transaction                                                                         | 60% | 48% | 35%  | 22% | 12% | -25%    |
| 2.1.2.3 | Retire outdated products/services                                               | Transaction                                                                         | 60% | 48% | 35%  | 22% | 12% | -25%    |
| 2.1.2.4 | Identify and refine performance indicators                                      | Knowledge                                                                           | 85% | 78% | 65%  | 52% | 42% | -20%    |
| 2.1.2.5 | Conduct post launch review                                                      | Knowledge                                                                           | 85% | 78% | 65%  | 52% | 42% | -20%    |
| 2.1.3   | Manage patents, copyrights, and regulatory requirements                         |  | 87% | 81% | 68%  | 57% | 47% | -19%    |
| 2.1.3.1 | Conduct mandatory and elective reviews                                          | Knowledge                                                                           | 85% | 78% | 65%  | 52% | 42% | -20%    |
| 2.1.3.2 | Review infringement of patents and copyrights                                   | Exception                                                                           | 95% | 92% | 82%  | 75% | 68% | -13%    |
| 2.1.3.3 | Determine patent and copyright needs                                            | Knowledge                                                                           | 85% | 78% | 65%  | 52% | 42% | -20%    |
| 2.1.3.4 | Define product technical documentation management requirements                  | Knowledge                                                                           | 85% | 78% | 65%  | 52% | 42% | -20%    |
| 2.1.3.5 | Manage regulatory requirements                                                  | Knowledge                                                                           | 85% | 78% | 65%  | 52% | 42% | -20%    |
| 2.1.4   | Manage product and service master data                                          |  | 64% | 53% | 40%  | 28% | 18% | -24%    |
| 2.1.4.1 | Manage materials master lists                                                   | Transaction                                                                         | 60% | 48% | 35%  | 22% | 12% | -25%    |
| 2.1.4.2 | Manage bills of material                                                        | Transaction                                                                         | 60% | 48% | 35%  | 22% | 12% | -25%    |
| 2.1.4.3 | Manage routings                                                                 | Transaction                                                                         | 60% | 48% | 35%  | 22% | 12% | -25%    |
| 2.1.4.4 | Manage specifications                                                           | Transaction                                                                         | 60% | 48% | 35%  | 22% | 12% | -25%    |
| 2.1.4.5 | Manage drawings                                                                 | Transaction                                                                         | 60% | 48% | 35%  | 22% | 12% | -25%    |
| 2.1.4.6 | Manage product/material classification                                          | Transaction                                                                         | 60% | 48% | 35%  | 22% | 12% | -25%    |
| 2.1.4.7 | Develop and maintain                                                            | Document                                                                            | 72% | 58% | 40%  | 28% | 18% | -32%    |

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| PCF          | L1-Domain / L2-Process Group / L3-Process / L4-Activity                                  | Pattern                                                                             | M1         | M2         | M3 ★       | M4         | M5         | Δ M1→M3     |
|--------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------|------------|------------|------------|------------|-------------|
|              | quality/inspection documents                                                             |                                                                                     |            |            |            |            |            |             |
| 2.1.4.8      | Maintain process specification data                                                      | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 2.1.4.9      | Manage traceability data                                                                 | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 2.1.4.10     | Review and approve data access requests                                                  | Decision                                                                            | 92%        | 88%        | 80%        | 72%        | 65%        | -12%        |
| <b>2.2</b>   | <b>Generate and define new product/service ideas · L2 AVG</b>                            | —                                                                                   | <b>83%</b> | <b>76%</b> | <b>63%</b> | <b>50%</b> | <b>40%</b> | <b>-20%</b> |
| <b>2.2.1</b> | <b>Perform discovery research</b>                                                        | Knowledge                                                                           | <b>85%</b> | <b>78%</b> | <b>65%</b> | <b>52%</b> | <b>42%</b> | <b>-20%</b> |
| 2.2.1.1      | Identify new technologies                                                                | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| 2.2.1.2      | Develop new technologies                                                                 | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| 2.2.1.3      | Assess feasibility of integrating new leading technologies into product/service concepts | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| <b>2.2.2</b> | <b>Generate new product/service concepts</b>                                             |    | <b>80%</b> | <b>72%</b> | <b>59%</b> | <b>46%</b> | <b>36%</b> | <b>-21%</b> |
| 2.2.2.1      | Gather new product/service ideas and requirements                                        | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 2.2.2.2      | Analyze new product/service ideas and requirements                                       | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| 2.2.2.3      | Evaluate new product/service inputs and requirements                                     | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| 2.2.2.4      | Formulate new product/service concepts                                                   | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| 2.2.2.5      | Identify potential improvements to existing products and services                        | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| <b>2.2.3</b> | <b>Define product/service development requirements</b>                                   | Knowledge                                                                           | <b>85%</b> | <b>78%</b> | <b>65%</b> | <b>52%</b> | <b>42%</b> | <b>-20%</b> |
| 2.2.3.1      | Define product/service requirements                                                      | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| 2.2.3.2      | Define post launch support model                                                         | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| 2.2.3.3      | Identify product/service bundling opportunities                                          | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| <b>2.3</b>   | <b>Develop products and services · L2 AVG</b>                                            | —                                                                                   | <b>77%</b> | <b>68%</b> | <b>55%</b> | <b>42%</b> | <b>33%</b> | <b>-22%</b> |
| <b>2.3.1</b> | <b>Design and prototype products and services</b>                                        |  | <b>80%</b> | <b>72%</b> | <b>59%</b> | <b>47%</b> | <b>37%</b> | <b>-21%</b> |
| 2.3.1.1      | Assign resources to product/service project                                              | Decision                                                                            | 92%        | 88%        | 80%        | 72%        | 65%        | -12%        |
| 2.3.1.2      | Prepare high-level business case and technical assessment                                | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| 2.3.1.3      | Develop product/service design specifications                                            | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| 2.3.1.4      | Develop user experience design specifications                                            | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| 2.3.1.5      | Provide warranty-related recommendations                                                 | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| 2.3.1.6      | Document design specifications                                                           | Document                                                                            | 72%        | 58%        | 40%        | 28%        | 18%        | -32%        |
| 2.3.1.7      | Conduct mandatory and elective external reviews                                          | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| 2.3.1.8      | Design products/services                                                                 | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| 2.3.1.9      | Build prototypes/proof of concepts                                                       | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 2.3.1.10     | Develop and test prototype production and/or service delivery process                    | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 2.3.1.11     | Eliminate quality and reliability problems                                               | Exception                                                                           | 95%        | 92%        | 82%        | 75%        | 68%        | -13%        |
| 2.3.1.12     | Conduct in-house product/service testing and evaluate feasibility                        | Transaction                                                                         | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 2.3.1.13     | Identify design/development performance indicators                                       | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| 2.3.1.14     | Collaborate on design with suppliers and external partners                               | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| <b>2.3.2</b> | <b>Test market for new or revised products and services</b>                              |  | <b>79%</b> | <b>70%</b> | <b>58%</b> | <b>44%</b> | <b>34%</b> | <b>-21%</b> |
| 2.3.2.1      | Prepare detailed market study                                                            | Knowledge                                                                           | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |

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|--------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------|------------|------------|------------|------------|-------------|
| 2.3.2.2      | Conduct customer tests and interviews                                 | Transaction                                                                       | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 2.3.2.3      | Finalize product/service characteristics and business cases           | Knowledge                                                                         | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| 2.3.2.4      | Finalize technical requirements                                       | Knowledge                                                                         | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| <b>2.3.3</b> | <b>Prepare for production/service delivery</b>                        |  | <b>65%</b> | <b>54%</b> | <b>41%</b> | <b>28%</b> | <b>18%</b> | <b>-24%</b> |
| 2.3.3.1      | Design and obtain necessary capabilities/materials and equipment      | Transaction                                                                       | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 2.3.3.2      | Identify requirements for changes to manufacturing/delivery processes | Knowledge                                                                         | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| 2.3.3.3      | Request engineering/process change                                    | Transaction                                                                       | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 2.3.3.4      | Install and validate production/service delivery process              | Transaction                                                                       | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 2.3.3.5      | Validate launch procedures                                            | Transaction                                                                       | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| <b>2.3.4</b> | <b>Define warranty claims</b>                                         |  | <b>76%</b> | <b>66%</b> | <b>52%</b> | <b>40%</b> | <b>31%</b> | <b>-24%</b> |
| 2.3.4.1      | Determine and document warranty policies                              | Document                                                                          | 72%        | 58%        | 40%        | 28%        | 18%        | -32%        |
| 2.3.4.2      | Create and manage warranty rules/claim codes for products             | Transaction                                                                       | 60%        | 48%        | 35%        | 22%        | 12%        | -25%        |
| 2.3.4.3      | Agree on warranty responsibilities with suppliers                     | Decision                                                                          | 92%        | 88%        | 80%        | 72%        | 65%        | -12%        |
| 2.3.4.4      | Define warranty related offerings for customers                       | Knowledge                                                                         | 85%        | 78%        | 65%        | 52%        | 42%        | -20%        |
| 2.3.4.5      | Communicate warranty policies and offerings                           | Document                                                                          | 72%        | 58%        | 40%        | 28%        | 18%        | -32%        |
| <b>2.3.5</b> | <b>Develop recall strategy</b>                                        | Knowledge                                                                         | <b>85%</b> | <b>78%</b> | <b>65%</b> | <b>52%</b> | <b>42%</b> | <b>-20%</b> |

■ Decision
 ■ Knowledge
 ■ Document
 ■ Transaction
 ■ Exception

Full L1→L2→L3→L4 hierarchy · Mixed-pattern rows show an equal-weighted rollup of their real L4 children · © Timothy P. King & Claude (Anthropic) 2026